

Homework #1 - Answers

1. Consider a disk with a sector size 200 bytes, 400 tracks per surface, 100 sectors per track, 4 double-sided platters, and average seek time of 16 milliseconds.

a) What is the capacity of a platter?

$$\text{Capacity of a platter} = 2 * 400 * 100 * 200 \text{ bytes}$$

b) What is the capacity of a track?

$$\text{Capacity of a track} = 100 * 200 \text{ bytes}$$

c) How many cylinders does the disk have?

$$\# \text{ of cylinders} = 400$$

d) What is the capacity of a cylinder?

$$\text{Capacity of a cylinder} = 2 * 4 * 100 * 200 \text{ bytes}$$

e) If the disk platters rotate at 2500 rpm, what is the average rotational delay?

$$\text{Max rotational delay} = \frac{1}{2500} * 60 * 1000 = 24 \text{ milliseconds}$$

$$\text{Avg rotational delay} = \frac{1}{2} * 24 = 12 \text{ milliseconds}$$

f) What is the block transfer time if one block contains 5 sectors?

$$1 \text{ block has 5 sectors, so 1 track has } \frac{100}{5} = 20 \text{ blocks}$$

$$\text{Therefore, btt} = \frac{24}{20} = 1.2 \text{ milliseconds}$$

2. Given the following $s = 16$ milliseconds, $r = 8.3$ milliseconds, $btt = 0.84$ milliseconds, $B = 2600$ bytes. A file having 80000 records and $R=400$ bytes is stored on this disk. Answer the following questions:

a) If this file is a pile file, calculate time to fetch a certain record.

Each block (page) can store $\frac{2600}{400} = 6$ records. So, the file has $\frac{80000}{6} = 13334$ blocks.

If it is a pile file, time to fetch one record is as follows:

$$T_F = s + r + \frac{b}{2} * btt = 16 + 8.3 + \frac{13334}{2} * 0.84 \cong 5624.58 \text{ milliseconds} \\ \cong 5.6 \text{ seconds}$$

b) If this file is a sorted sequential file with no overflow area, calculate time to fetch a certain record.

$$T_F = (\log_2 b) * (s + r + btt) = (\log_2 13334) * (16 + 8.3 + 0.84) \\ \cong 344.53 \text{ milliseconds} \cong 0.344 \text{ seconds}$$

c) If this file is a sorted sequential file such that 1/4 of the blocks are in the overflow area, calculate time to fetch a certain record.

$$\# \text{ of pages in the overflow area} = y = \frac{13334}{4} \cong 3334$$

$$\# \text{ of pages in the sorted area} = x = 13334 - 3334 = 10000$$

$$T_F = (\log_2 x) * (s + r + btt) + s + r + \frac{y}{2} * btt \\ = (\log_2 10000) * (16 + 8.3 + 0.84) + 16 + 8.3 + \frac{3334}{2} * 0.84 \\ \cong 1758.67 \text{ milliseconds} \cong 1.76 \text{ seconds}$$

d) If this file is an indexed file with a primary key index, calculate time to fetch a certain record given its primary key assuming that the whole primary key index can fit in main memory.

To find a record, firstly, binary search is applied to the primary key index to find the address of the record. As this search is done in memory, there is disk access cost of this operation. After that, 1 disk access is done to get the record.

$$T_F = s + r + btt = 16 + 8.3 + 0.84 = 25.14 \text{ milliseconds} \cong 0.025 \text{ seconds}$$

3. Assume that you try to estimate time required for batch update for a banking system. The master file has 100000 bank account information and each bank account record occupies 300 bytes. The transaction file contains 60000 transactions each occupies 300 bytes. Assume that no record is allowed to span more than one block, and 30% of the transactions are “open new account”, 20% of the transactions are “close an account”, and the others are “update” operations. The available memory for this operation is 3 buffer pages. Calculate a timing estimate for this batch update assuming that both the master and the transaction

files are sorted with respect to the account number. Use $s = 16$ milliseconds, $r = 8.3$ milliseconds, $btt = 0.84$ milliseconds, $B = 2000$ bytes.

1 page can store $\frac{2000}{300} = 6$ records. Therefore,

the master file has $\frac{100000}{6} = 16667$ pages

the transaction file has $\frac{60000}{6} = 10000$ pages

the output file has $100000 + 60000 * \frac{30}{100} - 60000 * \frac{20}{100} = 106000$ records, so

the output file has $\frac{106000}{6} = 17667$ pages.

3 buffer pages are available in memory for the batch update operation. This means that, 1 buffer page is used to read 1 block from the master file, 1 buffer page is used to read 1 block from the transaction file, 1 buffer page is used to write the output records. So, all disk accesses are random. As the master and transaction files are sorted, we read master and transaction files only one times, and write the output file.

Time to read the master file = $16667 * (16 + 8.3 + 0.84) = 419008.38$ milliseconds

Time to read the transaction file = $10000 * (16 + 8.3 + 0.84) = 251400$ milliseconds

Time to write the output file = $17667 * (16 + 8.3 + 0.84) = 444148.38$ milliseconds

Total time = 1114556.76 milliseconds $\approx$ 18.6 minutes